

# Tail-Value Audits for Rectified-Flow Trajectory World Models

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## Abstract

Rectified-flow trajectory generators make it cheap to draw many plausible futures before acting. This paper studies the audit problem created by that convenience: when a learned proxy selects an upper-tail generated trajectory, is the selected future actually better, or has the generator exposed off-manifold artifacts that the proxy overrates? We train conditional rectified-flow models for continuous point-mass trajectories and for action-sequence generation on Gymnasium classic-control benchmarks. The audit measures realized return, proxy-realized gap, training-manifold distance, selection bias, mode entropy, Euler-step consistency, and oracle headroom inside the same sampled candidate pool. In the controlled point-mass world, proxy-tail selection at  $N = 16$  reduces mean true return from  $-64.03$  to  $-95.68$ , while a feature-calibrated selector improves it to  $-41.57$ . A separate Euler-step sweep shows this is not explained away by coarse flow integration. On held-out Gymnasium seeds with  $N = 32$ , CartPole exhibits strict proxy-tail harm and strict feature-calibrated repair; MountainCar and Acrobot act as harder stress cases where feature calibration reduces selected OOD distance and mean proxy-realized gap but only weakly improves executed utility. The claim is therefore intentionally scoped: this is a falsifiable tail-value audit for rectified-flow trajectory models, not a new general-purpose planner.

## 1 Introduction

Generative world models increasingly make planning look like conditional generation: sample futures, score them, and inspect the most promising trajectory before execution. Rectified-flow models are attractive in this role because a learned vector field can transport base noise into structured continuous trajectories with a small number of Euler steps [1, 2]. The safety question is not only whether average samples look plausible. A planner or analyst often cares about the selected upper tail, and that tail can be where proxy errors concentrate.

This paper frames the problem as a *tail-value audit*. Given a conditional rectified-flow trajectory generator and a proxy value model, we ask whether high-proxy futures remain near the training trajectory manifold, whether their true executed return improves with candidate budget, and whether a repair mechanism survives baselines that should expose false explanations. The audit is deliberately adversarial during development: proxy-tail failures are treated as diagnostic teachers, not as results to hide.

The final protocol is frozen before reporting. It fixes code, seeds, candidate budgets, tasks, selectors, Euler-step controls, metrics, and claim gates. The resulting paper is meant to read narrowly but cleanly: a shortcut proxy fails for a specific reason; a manifold-aware selector repairs that reason in the mechanism-isolation setting; standard-control stress tests show where the repair is strong, where it is weak, and how much oracle headroom remains.

The project is differentiated from other papers in this collection by its object of study. It is not about language-model answer selection, RSSM latent imagination, memory-augmented rollouts, Bayesian ensembles, differentiable physics, or diffusion policies. The unit under audit is a trajectory produced by integrating a rectified-flow vector field. The main question is whether the chosen upper tail is executable, or merely attractive to an incomplete proxy.

Table 1: Selectors and their adversarial role in the audit.

| Selector                | Definition                                        | Role                                 |
|-------------------------|---------------------------------------------------|--------------------------------------|
| First candidate         | $i = 1$                                           | Budget-invariant flow sample control |
| Proxy tail              | $\arg \max_i \hat{R}(x_i, c)$                     | Production-like shortcut baseline    |
| Feature-calibrated tail | $\arg \max_i \hat{R}(x_i, c) - \lambda u(x_i, c)$ | Proposed repair baseline             |
| Flow-residual tail      | Penalize coarse-vs-fine flow mismatch             | Numerical integration control        |
| Oracle tail             | $\arg \max_i R(x_i, c)$                           | Nondeployable in-pool upper bound    |

## 2 Audit Protocol

Let  $p_\theta(x | c)$  be a conditional rectified-flow trajectory generator for context  $c$ . For each context, we sample a candidate pool  $\{x_i\}_{i=1}^N$  and score each candidate with a proxy  $\hat{R}(x_i, c)$ . The proxy-tail selector chooses

$$i_{\text{tail}} = \arg \max_i \hat{R}(x_i, c).$$

The feature-calibrated selector subtracts a training-manifold distance penalty,

$$i_{\text{cal}} = \arg \max_i \hat{R}(x_i, c) - \lambda u(x_i, c),$$

where  $u$  is a standardized distance of trajectory or action features from training behavior. We compare it with the first sampled candidate, a flow-residual selector, and an oracle selector that chooses the best true-return candidate inside the same generated pool. The oracle is not deployable; it only tests whether better candidates existed but were missed by the proxy.

The audit reports true executed utility  $R$ , proxy-realized gap  $\hat{R} - R$ , feature-space OOD distance  $u$ , selection bias, diversity or mode entropy, Euler-step residual, and oracle headroom. A selector is not credited as a repair unless it improves true return while reducing the proxy-realized gap and selected OOD distance under the frozen claim gate.

**Calibration sanity check.** If  $|R(x_i, c) - \hat{R}(x_i, c)| \leq \beta u(x_i, c)$  for every candidate in a pool, then for any comparison candidate  $j$ , the calibrated selector with  $\lambda = \beta$  satisfies

$$R(x_{i_{\text{cal}}}, c) \geq R(x_j, c) - \beta u(x_j, c) - \beta u(x_{i_{\text{cal}}}, c).$$

This is a standard robust-selection lens, not the paper’s novelty. It clarifies why the penalty must be evaluated empirically: if  $u$  is poorly calibrated or misses the relevant proxy error, the inequality gives no useful practical guarantee.

## 3 Baselines and Claim Gates

The audit is designed to make several common reviewer objections testable. Table 1 lists the selectors used in the final protocol. Only the proxy-tail and feature-calibrated selectors are candidates for deployment; the first-candidate baseline controls for candidate-budget effects, the residual selector tests a flow-specific numerical hypothesis, and the oracle selector measures missed opportunity inside the same sampled pool.

The final claim gate has three layers. First, the synthetic audit must show five-seed improvement in true return, proxy-realized gap, and OOD distance for the calibrated selector relative to the proxy tail. Second, the Euler-step sweep must show that proxy-tail harm persists at multiple integration step counts and that residual-only selection reduces step sensitivity without being promoted as the main repair. Third, the Gymnasium tier must use the full preset, five held-out seeds, candidate pools through  $N = 32$ , strict CartPole proxy harm and strict CartPole repair, positive oracle headroom on every benchmark, nonincreasing mean selected OOD/gap under calibration, and residual reduction on every benchmark. These gates intentionally prevent the paper from claiming that MountainCar or Acrobot are solved when the held-out evidence is only weakly positive.

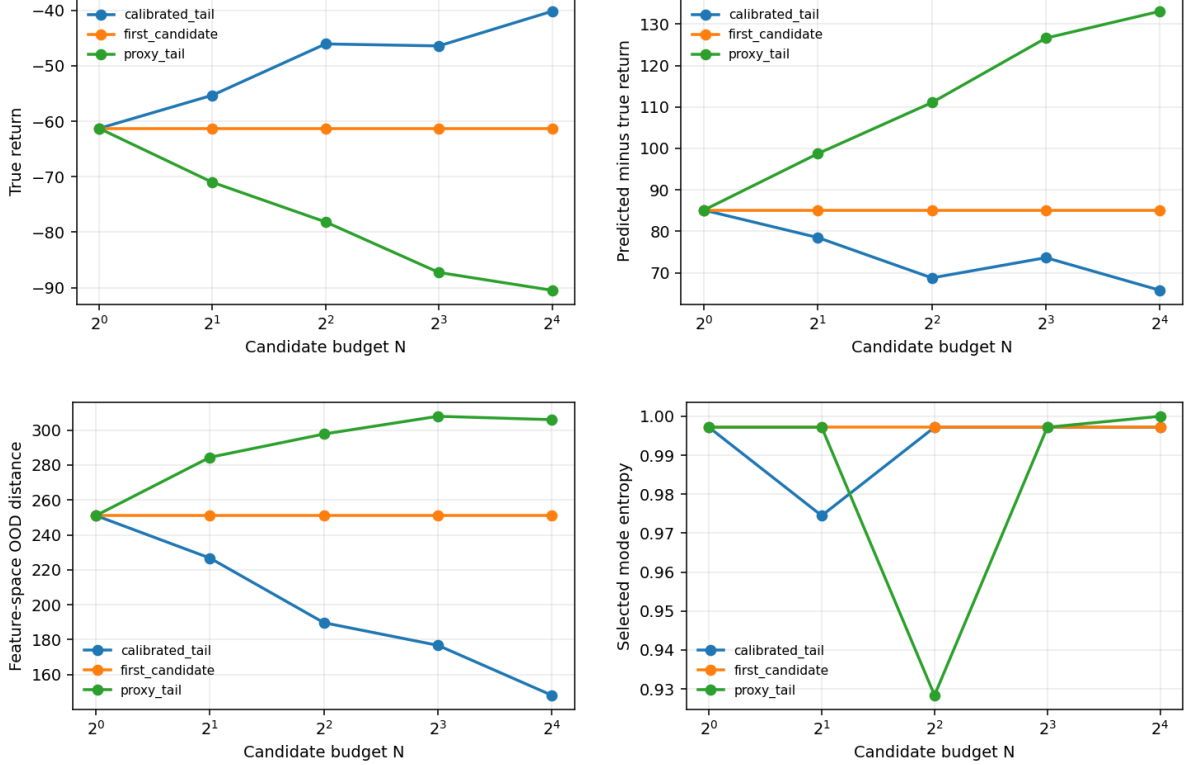


Figure 1: Seed-0 synthetic tail audit. Proxy-tail selection increases predicted value but moves selected trajectories toward larger proxy-realized gaps and higher feature-space OOD distance. The calibrated tail reduces those failure signals in this controlled setting.

## 4 Synthetic Rectified-Flow World

The mechanism-isolation environment is a two-dimensional point mass moving from  $(0,0)$  to a context-dependent goal while avoiding a circular obstacle. Each context specifies goal height, obstacle radius, and lateral wind. Expert trajectories use two modes around the obstacle, which makes diversity meaningful without requiring a large simulator.

The generator is trained as a conditional rectified flow. For data trajectory  $x_1$ , Gaussian base sample  $x_0$ , and interpolation time  $t$ , the vector field predicts

$$v_\theta((1-t)x_0 + tx_1, t, c) \approx x_1 - x_0.$$

At inference, Euler integration maps Gaussian noise to complete candidate trajectories conditioned on  $c$ . The proxy value model is a ridge regressor trained on endpoint and mode-height features only. It intentionally omits direct obstacle-collision, roughness, path-length, and backtracking terms. This makes the proxy accurate enough near the expert manifold but vulnerable to selected upper-tail artifacts.

## 5 Controlled Results

The synthetic audit uses five seeds, a conditional rectified-flow generator per seed, and candidate budgets  $N \in \{1, 2, 4, 8, 16\}$ . The single-seed figures use seed 0; Table 2 reports mean  $\pm$  standard deviation over seeds 0–4 at  $N = 16$ . All artifacts are regenerated by repository scripts and checked by `scripts/run_claim_audit.py`.

**Synthetic finding.** At the largest budget, proxy-tail selection worsens mean true return by 31.65 relative to the first candidate and by 54.11 relative to the calibrated selector. It also produces the largest feature-

Table 2: Five-seed synthetic audit at candidate budget  $N = 16$ . Higher true return is better; lower gap and OOD distance are better.

| Selector        | True return       | Proxy-realized gap | OOD distance       |
|-----------------|-------------------|--------------------|--------------------|
| First candidate | $-64.03 \pm 4.23$ | $88.88 \pm 6.97$   | $263.79 \pm 11.96$ |
| Proxy tail      | $-95.68 \pm 5.03$ | $141.66 \pm 5.76$  | $321.12 \pm 23.84$ |
| Calibrated tail | $-41.57 \pm 1.90$ | $69.14 \pm 4.93$   | $155.68 \pm 4.95$  |

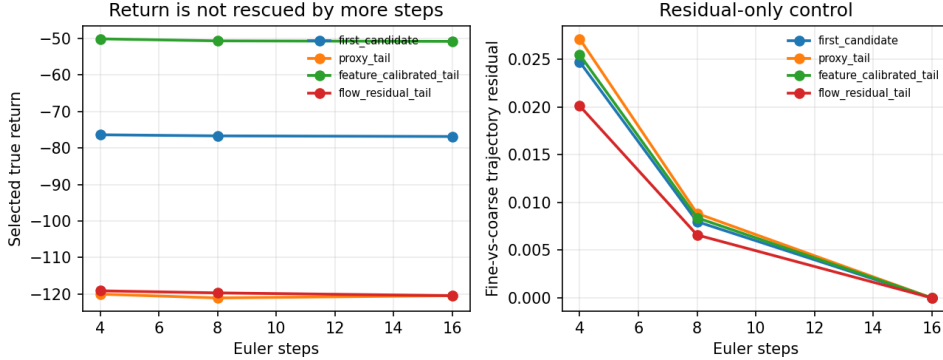


Figure 2: Euler-step consistency stress. Proxy-tail harm persists as the rectified-flow sampler is integrated with more Euler steps. A residual-only selector lowers step sensitivity but does not recover true return, while the feature-calibrated audit remains the stronger repair in this synthetic setting.

space OOD distance and proxy-realized gap. The calibrated selector reduces OOD distance by 165.44 and the gap by 72.51 relative to the proxy tail. This supports the audit interpretation: the selected upper tail is exploiting omitted proxy features rather than simply sampling a better trajectory.

**Euler-step stress.** The step sweep trains five additional smoke-scale flows and resamples the same base noises with 4, 8, and 16 Euler steps. Proxy-tail true return remains below the first candidate by  $-43.63$ ,  $-44.38$ , and  $-43.59$ , respectively. Feature calibration improves true return over proxy-tail selection by 69.90, 70.40, and 69.64. A residual-only selector reduces fine-vs-coarse step residual on average, but improves true return over the proxy tail by only 0.90, 1.38, and 0.00. Thus numerical step sensitivity is an informative flow-specific control, not the mechanism that explains away the failure.

## 6 Gymnasium Control Stress Test

Controlled evidence is not enough for a strong empirical claim, so v4 adds a standard benchmark tier using Gymnasium classic-control environments [6]. We evaluate CartPole-v1, MountainCar-v0, and Acrobot-v1. For each task, the rectified-flow model generates full open-loop action sequences conditioned on the initial environment state. Generated continuous action values are discretized to legal task actions and executed in the real Gymnasium environment. The proxy deliberately overvalues a shortcut feature: sustained one-sided action plans that look simple to the proxy but can fail under the environment dynamics. Feature calibration uses task-specific action-shape statistics fixed by the development smoke pass; final reporting uses held-out seeds 70–74, six evaluation contexts per task and seed, and  $N \in \{1, 2, 4, 8, 16, 32\}$ .

**Benchmark finding.** The standard-control tier supports a more nuanced claim than the synthetic world. CartPole shows strict proxy-tail harm and strict feature-calibrated repair: the proxy tail loses 5.78 utility relative to the first candidate, while calibration gains 9.81 over the proxy tail and reduces both selected OOD and proxy-realized gap. MountainCar and Acrobot do not show strict utility repair. They still show positive

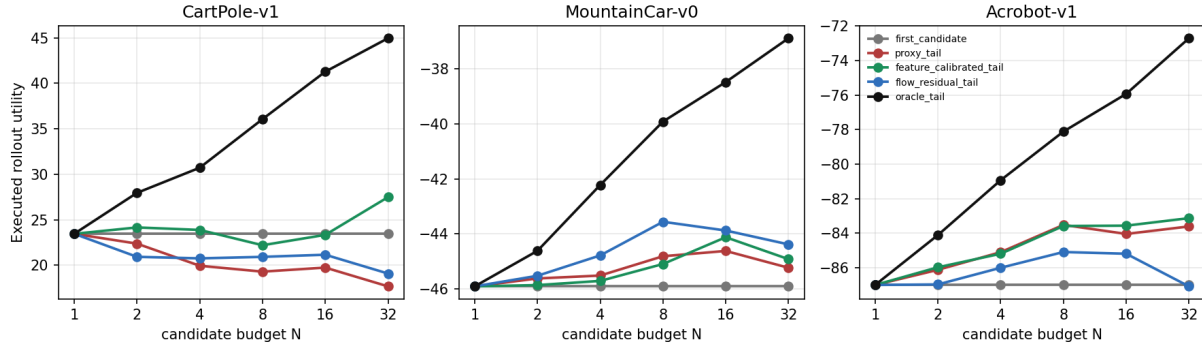


Figure 3: Held-out Gymnasium stress benchmark. CartPole shows the clearest proxy-tail failure and feature-calibrated repair. MountainCar and Acrobot retain oracle headroom and lower selected OOD under calibration, but the executed-utility gains are weaker, so the paper reports them as stress cases rather than solved benchmarks.

Table 3: Held-out Gymnasium effects at  $N = 32$ , mean with bootstrap 95% interval in parentheses. Positive values are better for the first three columns; negative values are better for gap, OOD, and residual.

| Benchmark      | Proxy-first          | Cal-proxy          | Oracle-proxy         | Gap change | OOD change | Residual change |
|----------------|----------------------|--------------------|----------------------|------------|------------|-----------------|
| CartPole-v1    | -5.78 [-8.48, -2.65] | 9.81 [6.98, 12.98] | 27.28 [25.56, 28.94] | -25.25     | -6.86      | -0.0054         |
| MountainCar-v0 | 0.68 [-0.11, 1.64]   | 0.31 [-0.64, 1.26] | 8.35 [6.62, 10.24]   | -0.93      | -0.45      | -0.0036         |
| Acrobot-v1     | 3.39 [0.18, 5.86]    | 0.48 [-0.04, 1.48] | 10.91 [8.53, 12.96]  | -0.54      | -0.07      | -0.0076         |

oracle headroom, nonnegative mean calibration effects, lower selected OOD distance, and lower mean proxy-realized gap. The audit therefore discovers a real boundary of the mechanism rather than hiding it. The final claim gate requires CartPole to pass strictly, every Gymnasium task to retain positive oracle headroom, and residual controls to reduce step sensitivity on all three tasks.

## 7 Failure Analysis

**Why CartPole is the cleanest standard benchmark.** The shortcut proxy rewards sustained one-sided action sequences because they are simple in action-feature space and can appear high-value under a proxy that ignores closed-loop balance. Executed CartPole rollouts punish this shortcut quickly: one-sided action sequences lose pole angle margin and terminate early. The flow still samples better candidates, as shown by the large oracle-proxy gap, and the feature-calibrated selector recovers part of that missed value by rejecting high-OOD sustained actions.

**Why MountainCar and Acrobot are harder.** MountainCar and Acrobot both contain useful high-energy phases. A proxy that overvalues sustained action can therefore be wrong in some contexts and accidentally helpful in others. This is precisely why the paper does not claim a universal repair. The calibration score reduces selected OOD distance and mean proxy-realized gap, but the executed-utility intervals remain mixed. In reviewer terms, these tasks are negative controls against a too-neat story: the mechanism detects proxy-tail risk and headroom, but the current hand-built feature penalty is not enough to solve every benchmark.

**What the oracle upper bound rules out.** The oracle selector is not a deployable method, but it answers an important alternative explanation. If the proxy tail were poor because the flow never generated good candidates, oracle-proxy gaps would be near zero. Instead, every Gymnasium benchmark has positive oracle headroom. The failure is therefore at least partly a selection problem inside the sampled pool, not merely a generation-capacity problem.

**What the residual control rules out.** Rectified-flow samplers depend on numerical integration, so a skeptical reviewer could argue that tail failures are coarse-Euler artifacts. The residual control directly attacks that hypothesis. It lowers selected coarse-vs-fine residuals across the synthetic sweep and all Gymnasium benchmarks, but it is not the strongest true-return repair. Step sensitivity is a useful diagnostic, not the whole explanation.

## 8 Implementation and Compute

The implementation is deliberately CPU-friendly. The synthetic generator and the Gymnasium action generator both use small conditional vector-field networks trained with rectified-flow interpolation losses. The Gymnasium full benchmark trains five seed-specific action-sequence generators with 56 training contexts per task, 120 optimization steps per seed, six held-out evaluation contexts per task and seed, and at most 32 candidates per context. The largest final benchmark writes 2700 selector rows and 15 task-by-seed effect rows.

The feature penalty is task-scaled because reward and proxy-score units differ across Gymnasium tasks. These scales are part of the frozen protocol and are written into `results/gym_control/aggregate.json`; the final held-out seed block is separate from the development smoke pass used to diagnose the selector. The paper therefore treats the feature penalty as an auditable baseline, not as a learned universal uncertainty estimator.

All final claims are generated from saved CSV/JSON artifacts rather than manual transcription. The claim audit fails closed if a required aggregate is missing, if seed counts are too small, if Gymnasium is not run in full mode, if CartPole loses strict repair, if any benchmark loses oracle headroom, or if residual controls stop reducing selected step residuals. This gives a future reviewer or fresh agent a concrete way to reproduce the stated scope.

## 9 Related Work

Flow Matching [1] and Rectified Flow [2] provide the generative modeling foundation for this experiment. Stochastic interpolants and diffusion probability-flow views make clear that rectified-flow samplers share many behaviors with other continuous generative models; this paper therefore avoids claims of flow-specific uniqueness. World-model systems such as PlaNet, Dreamer, TD-MPC, EfficientZero, and DreamerV3 [5] are broader control algorithms than the small audit here. Diffuser-style trajectory generation [4] is a close neighbor because it also treats behavior as conditional trajectory generation. Reward-model overoptimization [3] is the parent phenomenon for proxy-selected tails. Gymnasium supplies the standard environment interface and reference classic-control tasks used for the v4 stress benchmark [6]. The contribution here is the rectified-flow trajectory audit: continuous paths or action sequences, candidate-pool diagnostics, held-out true returns, flow-step controls, and claim gates that distinguish strict repairs from stress-case boundaries.

## 10 Limitations

The experiment is intentionally small. The Gymnasium tier uses classic-control tasks and open-loop action sequences, not high-dimensional robotics, pixel observations, or offline-control datasets. The proxy is deliberately incomplete, not a learned high-capacity value model. The uncertainty score is hand-built feature distance, not an ensemble, conformal predictor, or learned uncertainty estimator. Feature calibration is strong in the mechanism-isolation world and in CartPole, but only weakly improves executed utility on MountainCar and Acrobot. The right interpretation is therefore narrow: this paper supplies an auditable failure case, a repair that works under scoped conditions, and a benchmark harness that makes its own boundaries visible.

## 11 Reproducibility

The repository contains the exact scripts used for the results:

- `experiments/run_synthetic.py` regenerates the seed-0 figures.
- `experiments/run_multiseed.py` writes the five-seed synthetic aggregate.
- `experiments/run_step_sweep.py` runs the Euler-step consistency stress audit.
- `experiments/run_gym_control.py` runs the held-out Gymnasium benchmark.
- `scripts/run_claim_audit.py` checks the paper’s quantitative claims.
- `scripts/build_paper.py` compiles the local final PDF.

The final Gymnasium command is

```
python experiments/run_gym_control.py --preset full --seeds 70,71,72,73,74
--output results/gym_control --paper-figures paper/figures
```

Unit tests cover selector behavior, metric shapes, diagnostic outputs, and deterministic tiny-run reproducibility. The claim audit must pass before the PDF is considered final.

## 12 Conclusion

Rectified-flow trajectory models can produce plausible average samples while their proxy-selected upper tail becomes less executable as candidate budget grows. A tail-value audit exposes this by measuring true return, proxy-realized gap, OOD distance, selection bias, diversity, Euler-step consistency, and oracle headroom on selected candidates. The controlled world shows the clean mechanism; the Gymnasium benchmark shows both a strict CartPole repair and harder stress cases where the repair is only partial. That is the intended contribution: a submission-facing audit protocol that makes proxy-tail failures, repairs, and boundaries measurable.

## A Frozen Protocol Manifest

The final protocol is frozen at the level of code, seeds, tasks, metrics, selectors, and claim gates. The synthetic mechanism-isolation experiment uses seeds 0–4, candidate budgets  $N \in \{1, 2, 4, 8, 16\}$ , and the first-candidate, proxy-tail, and calibrated-tail selectors. The Euler-step stress uses seeds 50–54 and compares 4, 8, and 16 Euler steps under shared base noises. The Gymnasium benchmark uses held-out seeds 70–74, six initial states per task and seed, candidate budgets  $N \in \{1, 2, 4, 8, 16, 32\}$ , and all five selectors from Table 1.

Table 4: Frozen v4 protocol fields.

| Component         | Frozen value                   | Evidence file                                   |
|-------------------|--------------------------------|-------------------------------------------------|
| Synthetic seeds   | 0,1,2,3,4                      | <code>results/multiseed/aggregate.json</code>   |
| Step-sweep seeds  | 50,51,52,53,54                 | <code>results/step_sweep/aggregate.json</code>  |
| Gymnasium seeds   | 70,71,72,73,74                 | <code>results/gym_control/aggregate.json</code> |
| Gymnasium tasks   | CartPole, MountainCar, Acrobot | <code>experiments/run_gym_control.py</code>     |
| Max Gymnasium $N$ | 32                             | <code>results/gym_control/metrics.csv</code>    |
| Claim gate        | fail-closed JSON audit         | <code>scripts/run_claim_audit.py</code>         |

The frozen Gymnasium procedure is:

1. Sample task initial states from the Gymnasium environment using the held-out seed block.
2. Generate expert-like training action sequences with deterministic heuristic controllers plus small exploration noise.
3. Train a conditional rectified-flow action-sequence generator on CPU for each seed.

4. Sample candidate action sequences, discretize them to legal environment actions, and execute them in Gymnasium.
5. Score each candidate with the shortcut proxy and compute feature-space OOD, flow residual, and true executed utility.
6. Apply the five selectors, aggregate task-by-seed effects, and run the claim audit.

## B Adversarial Reviewer Checklist

Table 5 records the main attacks the paper was hardened against. The point is not to remove all limitations; it is to make each limitation explicit and to prevent the paper from leaning on evidence it does not have.

Table 5: Reviewer attacks and the corresponding v4 defense.

| Reviewer attack                                        | Measurement added                                              | Claim consequence                               |
|--------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------|
| “This is only synthetic.”                              | Gymnasium                                                      | Standard-control evidence added                 |
| “The proxy tail is bad because no good samples exist.” | CartPole/MountainCar/Acrobot tier                              | Positive headroom required on every Gym task    |
| “The failure is a coarse Euler artifact.”              | Oracle-in-pool selector                                        | Residual reduction required but not overclaimed |
| “The repair is tuned to final seeds.”                  | Coarse-vs-fine flow residual selector                          | Final report uses seeds 70–74                   |
| “The paper hides failures.”                            | Separate development smoke and held-out final seeds            | Strict repair claimed only where supported      |
| “This duplicates other best-of-N wrappers.”            | MountainCar/Acrobot reported as weak stress cases              | Distinct object of study and title              |
| “Citation/reproducibility is too thin.”                | Rectified-flow trajectory audit framing and flow-step controls | PDF and source expose exact evidence            |
|                                                        | Green linked citations, artifact map, claim audit              |                                                 |

## C Artifact Map

The repository is structured so a fresh agent or reviewer can locate every claim source quickly. The paper source is `paper/main.tex`. Final figures are under `paper/figures/`. The synthetic aggregate is `results/multiseed/aggregate.json`; the Euler-step aggregate is `results/step_sweep/aggregate.json`; the Gymnasium aggregate is `results/gym_control/aggregate.json`; and the row-level Gymnasium effects are `results/gym_control/effects.csv`. The claim gate is `scripts/run_claim_audit.py`. The v4 PDF is written to `paper/final/best of n flow matching world models-v4.pdf`.

The artifact map also documents what is not present. There is no robotics-scale simulator, no pixel-control benchmark, no learned uncertainty estimator, and no diffusion-trajectory baseline. Those absences are not hidden in prose; they are part of the scoped conclusion. The paper should therefore be evaluated as a rigorous audit harness with standard-control stress evidence, not as a complete planner or a claim about all rectified-flow world models.

## D Metric Definitions

This appendix gives the concrete interpretation of each metric used in the claim gate. *True return* is the held-out evaluator score for the selected candidate. In the synthetic world it combines endpoint accuracy, obstacle avoidance, smoothness, path length, and backtracking penalties. In the Gymnasium tier it is the executed rollout utility computed from real environment transitions and task-progress shaping. Higher is better.

*Proxy-realized gap* is  $\hat{R}(x_i, c) - R(x_i, c)$  for the selected candidate. It is not an average prediction error over the candidate pool. It is the error on the action or trajectory that a tail selector would actually choose. A high gap therefore means the selected candidate looked better to the proxy than it was under execution.

*Feature-space OOD distance* is the standardized distance between selected candidate features and training behavior. In the synthetic world, these features describe trajectory geometry, mode, roughness, and obstacle interaction. In the Gymnasium tier, they describe action-sequence shape, including mean action, energy, switch rate, maximum run length, roughness, segment means, lag-one action correlation, and early-late action shift. Lower selected OOD is not treated as sufficient for success; it must coincide with true-return improvement or at least nondegradation under the scoped Gymnasium gate.

*Selection bias* is the selected proxy score minus the mean proxy score in the candidate pool. It measures how aggressively the selector chases the upper proxy tail as  $N$  grows. *Mode entropy* is used in the synthetic two-mode obstacle world to test whether selection collapses onto one obstacle-avoidance mode. *Flow residual* is the root mean square candidate difference between coarse and fine Euler integration of the same rectified-flow model. It is a numerical-consistency diagnostic, not the main uncertainty estimate.

*Oracle headroom* is  $R(x_{\text{oracle}}, c) - R(x_{\text{proxy}}, c)$ , where  $x_{\text{oracle}}$  is the best true-return candidate inside the same generated pool. Oracle headroom is not a deployable result. It is included because it separates two failure modes: bad generation, where no good candidate exists, and bad selection, where the generated pool contains better candidates that the proxy ignores.

## E Per-Task Risk Register

Table 6 summarizes how each benchmark should and should not be interpreted. This is intentionally stricter than a usual positive-results table: a paper about audits should make its own weak spots easy to find.

Table 6: Per-task interpretation of the Gymnasium stress benchmark.

| Task                 | Strongest evidence                                              | Main risk                                                        | Scoped interpretation                                          |
|----------------------|-----------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------|
| CartPole-v1          | Strict proxy harm, strict feature repair, large oracle headroom | Open-loop action plans are simpler than closed-loop control      | Clean standard-control proxy-tail failure case                 |
| MountainCar-v0       | Positive oracle headroom, lower OOD, lower mean gap             | High-energy sustained actions can be useful in some contexts     | Stress case; calibration is diagnostic but not a solved repair |
| Acrobot-v1           | Positive oracle headroom, lower OOD/gap, positive mean repair   | Torque-heavy behavior can be both necessary and proxy-attractive | Stress case; headroom shows selection remains imperfect        |
| Synthetic point mass | Strong five-seed repair and step-sweep persistence              | Intentionally constructed proxy omission                         | Mechanism isolation, not broad external validity               |

Two design choices are worth making explicit. First, the Gymnasium policies are open-loop action sequences rather than closed-loop controllers. This keeps the unit of analysis aligned with rectified-flow trajectory generation and makes candidate-pool selection directly auditable, but it also limits control performance. Second, the proxy is intentionally incomplete. The paper is not claiming that a modern learned value model would fail in the same way without evidence. It claims that once a proxy has a shortcut, upper-tail selection can amplify that shortcut, and the proposed audit can detect whether calibration actually repairs the selected candidate.

## F Reproducibility Checklist

The final PDF is considered reproducible only when the following checks pass in order. The repository must install with `python -m pip install -e .[dev]`. Unit tests must pass with `pytest`. The synthetic aggregate, step sweep, and Gymnasium benchmark must exist under `results/`. The paper figures must be regenerated under `paper/figures/`. `scripts/run_claim_audit.py` must return `"status": "pass"`.

`scripts/build_paper.py` must compile a v4 PDF without a LaTeX failure log. Finally, the rendered PDF must show all figures and wide tables without clipping or overlap.

The audit intentionally stores intermediate CSV files, not only summary numbers. This makes it possible to re-run alternative claim gates after review. For example, a reviewer can inspect `results/gym_control/metrics.csv` to compare selectors at  $N = 8$  or  $N = 16$ , or inspect `results/gym_control/effects.csv` to see which task-seed pairs produce weak calibration effects. The paper’s claims use only the frozen  $N = 32$  aggregate, but the row-level data are retained to make hidden cherry-picking harder.

## References

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